DETAILED FUNCTIONAL SCOPE AND COMMERCIAL PROPOSAL

Integrated HR and finance operations platform

A secure, searchable web platform for workforce records, document expiry control, invoice processing and purchase-document reconciliation - replacing spreadsheet-led operations in controlled phases.

HR onboarding | Document OCR | AI translation | Finance MVP | Purchase cases | Supabase | Cloudflare

Two modules, one controlled operational system

The platform centralises people, documents, invoices, purchase cases and operational alerts. The HR and Financial modules remain independently deliverable, permissioned and priced, while sharing one secure technical foundation.

Current HR	Email threads → Manual entry → Folders → Calendar memory
Future HR	Invite link → Guided form → OCR review → Profile + alerts
Current finance	Mailbox / paper → Excel → Manual pairing → Payment check
Future finance	Upload / email → OCR + translate → Purchase case → Dashboard

MODULE A

HR administration

- ✓ Employee and candidate master records
- ✓ Secure onboarding invitations
- ✓ Personal-data and document collection
- ✓ Expiry-date OCR and review
- ✓ Templates, generated PDFs and signed scans
- ✓ Missing-document and expiry dashboard

MODULE B

Financial and banking

- ✓ Manual and email document intake
- ✓ Invoice OCR and editable register
- ✓ English AI translation after processing
- ✓ Purchase cases with 4–5 related documents
- ✓ EKÁER finalisation when eligible
- ✓ Payment export, confirmation and bank reconciliation
- ✓ Financial dashboards and document search

Original preserved

Structured OCR data never replaces the source file.

Human confirmation

Uncertain fields and ambiguous matches enter review.

No Excel dependency

Exports remain available; the platform becomes the source of truth.

From invitation to a verified employee dossier

01 · CREATE
Admin creates a named record with email, language and onboarding type.

02 · INVITE
System sends a unique, expiring and revocable link.

03 · COMPLETE
Worker enters data and saves progress on mobile or desktop.

04 · UPLOAD
Required identity, permit and insurance documents are added.

05 · REVIEW
Admin confirms values, completeness and employment readiness.

Employee profile

IDENTITY & TRAVEL

- Legal and former name
- Birth details and mother's name
- Nationality / citizenship
- National or foreign personal ID
- Passport / travel document number
- Issuing country/authority, issue and expiry dates

CONTACT

- Email and telephone
- Permanent and current address
- Address card / registration identifier
- Emergency contact
- Preferred communication language

EMPLOYMENT & PAYROLL

- Position and work location
- Legal employer / customer assignment
- Start/end date and worker status
- Employee / worker ID
- Bank account / IBAN, if collected
- Contract reference and responsible owner

COMPLIANCE IDENTIFIERS

- Tax identification number / tax card
- TAJ / social security number
- Health insurance provider and member/policy ID
- Residence permit type, number and validity
- Work permit type, number and validity
- Visa type, number and expiry, where required

Invitation controls

- ✓ Token is stored as a hash and is single-purpose.
- ✓ Validity period is configurable; resend creates a new token.
- ✓ Optional email OTP before sensitive form access.
- ✓ Privacy notice acknowledgement and version are logged.
- ✓ Conditional sections depend on nationality, permit route and job type.
- ✓ Server-side validation is authoritative.
- ✓ Final submission locks critical fields until an audited reopen.

Lifecycle

`Draft` → `Sent` → `Opened` → `Submitted` → `Approved`

Duplicate control: possible duplicate people are proposed from email, name and confirmed document identifiers. An administrator decides whether to merge, link or retain separate records.

EXCEPTION	PLATFORM RESPONSE	ADMIN ACTION
Expired/revoked link	No personal data exposed	Issue a new invitation
Incomplete submission	Missing sections remain visible	Request completion or reopen
Unsupported/large file	Upload rejected with clear reason	Request supported scan format
OCR uncertainty	Value stays unconfirmed	Compare with original and approve

Expiry extraction, document history and actionable alerts

DOCUMENT TYPE	SUGGESTED EXTRACTED FIELDS	OPERATIONAL USE
Passport / travel document	Number, nationality, expiry date	Identity and expiry control
Residence permit	Permit type/number, valid from/to	Immigration readiness
Work permit	Reference, validity, employer limits	Eligibility checklist
Tax card / tax identifier	Identifier; validity where applicable	Payroll reference
TAJ / social security card	Identifier and validity where present	Coverage checklist
Insurance certificate	Provider, policy, expiry date	Renewal control
Employment contract / annex	Document type, parties, dates, signature state	Contract completeness

UPLOAD
Private original + checksum

CLASSIFY
User-selected type

EXTRACT
Value + confidence + source page

VALIDATE
Date and identifier rules

CONFIRM
Human-approved active record

OCR SAFEGUARDS

- ✓ Raw provider result retained under controlled policy
- ✓ Low confidence never auto-approved
- ✓ International date formats normalised
- ✓ Correction stores user and timestamp
- ✓ Provider adapter can be replaced

TEMPLATE WORKFLOW

- ✓ Astex supplies legally approved template
- ✓ Fields map to confirmed profile values
- ✓ Versioned printable PDF is generated
- ✓ Physical signature occurs outside platform
- ✓ Signed scan is uploaded to profile

EXPIRY WORKFLOW

- ✓ Example windows: 90/60/30/14/7 days
- ✓ Owner and severity shown on dashboard
- ✓ Email notifications are deduplicated
- ✓ Replacement preserves old document
- ✓ Overdue action requires resolution/reason

HR dashboard

12
EXPIRING SOON

5
MISSING DOCUMENTS

3
UNSIGNED CONTRACTS

2
OCR REVIEWS

Configurable requirement matrix: required fields and document types may vary by nationality, employment route, job category or customer assignment without creating a separate application per country.

Phase 1: OCR, document control and payment preparation

The first financial phase replaces the primary invoice spreadsheet without waiting for direct bank connectivity. Staff upload or email every supported invoice, review OCR fields, receive English AI translations of the invoice content, organise related scans, prepare bank payments and control status from one interface.

01 · INTAKE
Upload or email invoice and related documents.

02 · OCR
Invoice fields are proposed with confidence.

03 · REVIEW + TRANSLATE
Confirm extracted data and create the English AI translation.

04 · OPERATE
Link cases and complete eligible EKÁER actions.

05 · PAY
Generate transfer export and confirm submission.

Phase 1 functions

- ✓ Role-based finance workspace and audit trail
- ✓ Manual and email invoice intake
- ✓ Invoice OCR with side-by-side original preview
- ✓ Editable values and validation messages
- ✓ English AI translation after OCR review
- ✓ Original and translated text stored together
- ✓ Sort, filter, pagination and saved table views
- ✓ Purchase-case creation, matching and review
- ✓ Delivery note, EKÁER, closing and other files
- ✓ Eligible EKÁER finalisation through the API
- ✓ Bank-import payment file and confirmation dialog
- ✓ CSV/XLSX export with active filters

INITIAL INVOICE FIELDS

- Supplier name
- Supplier tax ID
- Invoice number
- Issue date
- Performance date
- Due date
- Order/reference
- Invoice subject
- Currency
- Net / VAT / gross
- Product or item names
- Extracted line descriptions
- Relevant content text
- Source language
- English AI translation
- OCR confidence

Final mapping is confirmed from representative Astex invoices before OCR acceptance testing.

Invoice register views

NEW / REVIEW
OCR waiting, failed or low confidence.

OPEN
Approved invoices not fully paid.

INCOMPLETE
Related required files are missing.

CLOSED
Complete and reconciled records.

Phase 1 handover outcome: finance staff can stop using Excel as the authoritative record and operate invoice OCR, English AI translation, related-document control, EKÁER completion, dashboards and payment preparation before the direct bank connection is enabled.

One purchase case connects every related document

PURCHASE CASE HEADER

A complete operational dossier

Supplier, order/reference, dates, currency, amount, delivered weight, owner, completeness state, payment state and exception state.

Invoice · Delivery note · EKÁER declaration · Closing document · Other evidence

Case behaviour

- ✓ Each file keeps its original, version and checksum.
- ✓ One file has one active primary case; reassignment is audited.
- ✓ Required document roles are configurable by purchase type.
- ✓ Completeness, consistency and payment are separate controls.
- ✓ Notes, extracted data, matches and payment events share one timeline.
- ✓ Nothing is silently deleted, merged or closed.

New → Processing → Incomplete → Ready → Paid → Closed

Explainable matching score

SIGNAL	NORMALISATION	ROLE IN SCORE
Invoice / order / reference number	Remove punctuation, spaces and case differences	Strong exact or near-exact key
Supplier	Tax ID first; controlled aliases for legal/trading names	Strong tax ID, medium name match
Dates	ISO format and configurable sequence/window	Supporting chronological signal
Amounts and currency	Decimal/currency normalisation and tolerance	Strong invoice/payment signal
Delivered weight	Convert kg/tonne and decimal formats	Strong logistics/EKÁER signal
Goods description	Normalised text and controlled synonym dictionary	Supporting signal and search text

HIGH CONFIDENCE

Auto-link

Unique identifiers plus consistent supplier, dates and amount/weight. Explanation is stored.

MEDIUM CONFIDENCE

Review ranked candidates

User sees agreements, conflicts and alternative purchase cases.

LOW CONFIDENCE

Remain unmatched

Document stays in the exception queue until manually resolved.

EKÁER API completion: when all mandatory declaration data is present and validated, and Astex provides valid NAV EKÁER API access, the platform can create, modify and finalise (close) the EKÁER declaration through the API. Missing or rejected data remains in review; the request, response and finalisation timestamp are retained in the audit trail.

Quality safeguard: thresholds are configurable per document type and validated against a labelled historical sample. Ambiguous documents are never auto-linked only because one weak field happens to match.

Email-based reconciliation now, direct bank API later

INBOUND MAILBOX

- ✓ Dedicated Cloudflare-routed address
- ✓ MIME parsing and attachment extraction
- ✓ Sender, size and file-type validation
- ✓ Message-ID and checksum deduplication
- ✓ Unsupported items routed to review
- ✓ Same OCR pipeline as manual upload

NAV ONLINE INVOICE

- ✓ Dedicated Astex technical user
- ✓ Credentials kept as server-side secrets
- ✓ Test and production environments separated
- ✓ Invoice query and identity reconciliation
- ✓ Pagination, failures and retries logged
- ✓ NAV record does not replace source invoice

PAYMENT PREPARATION

- ✓ Select approved, unpaid invoices
- ✓ Validate beneficiary, account and amount
- ✓ Generate agreed bank-import file
- ✓ Upload and authorise in the bank interface
- ✓ Ask whether transfers were submitted
- ✓ Record confirmation, user, time and batch

Automation flow

Invoice	Receive → OCR → Review + translate → Register
EKÁER	Check required data → Create / modify → Finalise via API → Store response
Payment file	Select invoices → Export bank file → Authorise at bank → Confirm in platform
Bank email	Receive notice → Extract payment → Propose match → Confirm state

Included email alternative

- ✓ One agreed bank-notification email format.
- ✓ Extract reference, amount, currency and beneficiary.
- ✓ Propose invoice or payment-batch match.
- ✓ Human confirms or corrects the payment state.
- ✓ Unmatched messages remain visible for review.

Operational dashboards instead of disconnected spreadsheets

Financial dashboard

€ –	– %
MONTHLY SPEND	PAID ON TIME
–	–
INCOMPLETE CASES	OCR / MATCH REVIEW

- ✓ Monthly cost by supplier, category, currency and payment state
- ✓ Outstanding and overdue invoices
- ✓ Missing-document workload and ageing
- ✓ Supplier purchase history and recurring goods
- ✓ Filtered exports for reconciliation and management

HR dashboard

–	–
EXPIRY ACTIONS	INCOMPLETE PROFILES
–	–
UNSIGNED DOCUMENTS	NEW SUBMISSIONS

- ✓ Severity and due-date action queue
- ✓ Assigned HR owner and employee grouping
- ✓ Profile completeness by onboarding schema
- ✓ Nationality, customer, location and status filters
- ✓ Document renewal and contract follow-up

BASE SEARCH - INCLUDED

- ✓ Invoice/reference numbers, supplier and tax ID
- ✓ Dates, amounts, currency and payment status
- ✓ OCR line descriptions and document text
- ✓ English AI translations of invoice content
- ✓ Exact, prefix and Postgres full-text queries
- ✓ Controlled aliases and recurring-goods synonyms

Example: search "forklift" and return matching invoices even when the original invoice used another language.

AI TRANSLATION - INCLUDED

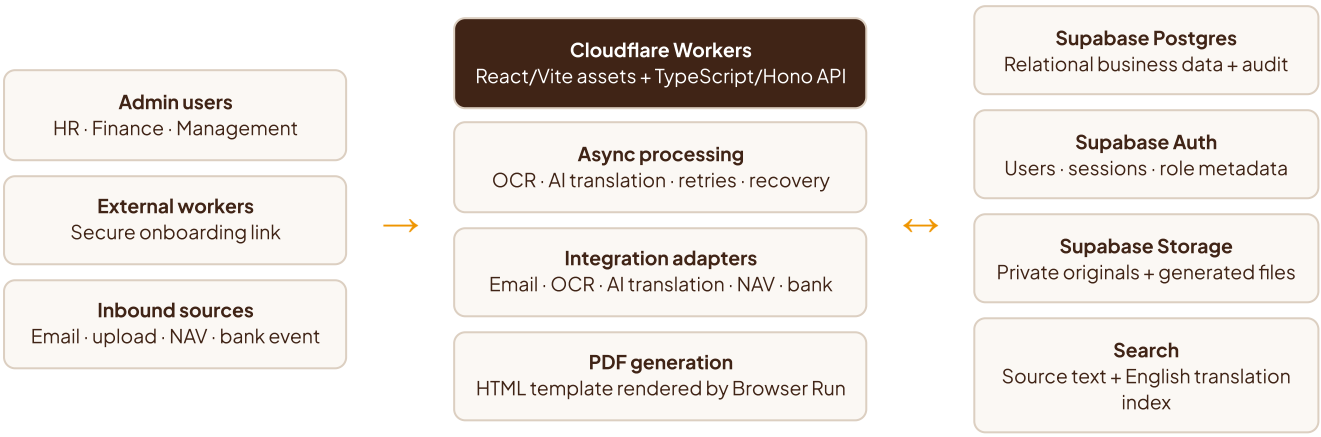
English invoice content after OCR

After OCR review, AI translates the invoice subject, product or item names, line descriptions and relevant extracted content into English. Source text and translation remain linked and searchable.

Astex pays all third-party AI model, translation API, subscription and usage charges. Conversational or semantic AI search is a separate extension.

Excel transition: existing sheets are mapped and imported after data cleanup. During parallel operation, exports remain available. After UAT, the database becomes the authoritative record and Excel is retained only for controlled exchange.

Supabase data core with a Cloudflare application edge



SHARED ENTITIES

- organisations
- users / memberships / roles
- files / file versions
- OCR jobs / extracted fields
- notifications
- audit events

HR ENTITIES

- people / employee profiles
- invitations / submissions
- form schemas / answers
- document requirements
- person documents / expiries
- templates / generated files

FINANCE ENTITIES

- suppliers / aliases
- purchase cases
- invoices / invoice lines
- finance documents / matches
- payment events
- NAV runs / inbound emails

Key technical decisions

- ✓ Cloudflare Workers Static Assets for the SPA and API deployment unit
- ✓ Cloudflare Queue for OCR, AI translation and external integration work
- ✓ Supabase selected in an appropriate EU region
- ✓ Private object buckets with signed short-lived downloads
- ✓ Organisation identifier on every business row
- ✓ Unique checksum/message/business constraints for idempotency
- ✓ GIN full-text and b-tree operational indexes
- ✓ Production and staging environments separated

Connection design: API requests use controlled Supabase interfaces or a carefully configured Postgres connection layer. Secrets and service credentials remain server-side and are never included in browser bundles.

Explicit access controls for personal and financial data

ROLE	HR	FINANCE	ADMINISTRATION
Platform owner	Full	Full	Users, settings and audit
HR administrator	Create/edit/export	No access by default	HR forms, templates, reminders
Finance administrator	No access by default	Create/edit/reconcile	Finance rules and integrations
Invitee	Own invitation only	None	No administration

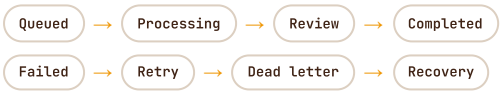
APPLICATION SECURITY

- ✓ Supabase Auth and optional MFA for administrators
- ✓ Explicit table grants and Row Level Security on exposed tables
- ✓ Authorisation uses server-controlled app metadata, not user-editable metadata
- ✓ Private storage policies and time-limited file access
- ✓ File type/size validation, rate limits and malware-scan hook
- ✓ Server-only NAV, OCR, AI translation and database secrets

DATA GOVERNANCE

- ✓ Purpose and legal basis confirmed by Astex
- ✓ Privacy notice and processor agreements before production
- ✓ Configurable retention by record/document category
- ✓ Export/deletion subject to statutory retention duties
- ✓ EU region preference and subprocessor inventory
- ✓ DPIA recommended for identity and workforce documents

Reliable background processing



- ✓ At-least-once delivery handled by idempotent consumers
- ✓ Per-message error classification and bounded retries
- ✓ Failed jobs have visible owner and recovery action

Audit and observability

- ✓ Actor, role, organisation and request correlation ID
- ✓ Before/after delta for sensitive changes
- ✓ Source: manual, OCR, AI translation, email, NAV, bank or scheduler
- ✓ Reason required for overrides, reassignment and deletion
- ✓ Alerts for OCR, translation, email, NAV and notification failures
- ✓ Backup configuration and periodic restore-process verification

Compliance boundary: technical controls support GDPR-conscious operation but do not replace Astex's legal assessment, privacy notices, retention rules or internal permission governance.

Phased implementation with early operational value

Stage 0	**Discovery and shared foundation** - samples, fields, roles, environments, security baseline and UI shell.	1
Finance 1A	**OCR and translation register** - upload/email intake, OCR review, English AI translation, register, cases and scans.	
Finance 1B	**Operational automation** - matching, NAV, eligible EKÁER finalisation, bank email, payment file and dashboard.	
HR 1	**HR onboarding** - profiles, invitations, forms, uploads, expiry OCR and administrative review.	
HR 2	**HR automation and handover** - alerts, dashboard, one template, generated PDF, UAT fixes and training.	
Optional 2	**Direct bank API** - separate discovery and quote only if Astex secures supported bank access.	

Minimum acceptance outcomes

- ✓ Role separation and organisation-isolation tests pass.
- ✓ HR invitation can be issued, revoked, completed and approved.
- ✓ Expiry values retain confidence, original evidence and human correction.
- ✓ Configured reminders create no duplicate actions.
- ✓ Supplied HR template generates a printable versioned PDF.
- ✓ Supported invoices enter the register through OCR review.
- ✓ Invoice subject, products, line descriptions and relevant content receive a stored English AI translation.
- ✓ Other document roles attach to a purchase case.
- ✓ Ambiguous matching produces ranked review, not silent linking.
- ✓ Agreed email/NAV test scenarios process without duplication.
- ✓ Dashboard totals reconcile with the approved UAT dataset.

Testing and handover

TEST LAYERS

- Parser/scoring unit tests
- Database and integration tests
- RLS allow/deny coverage
- End-to-end browser workflows
- Retry, dead-letter and restore checks

UAT PROTOCOL

Astex nominates one owner per module, supplies representative anonymised/redacted samples and responds within five business days per milestone. Defects are separated from new requests.

OCR accuracy: a numeric precision commitment can only be agreed after document languages, layouts, scan quality and a statistically useful labelled sample are reviewed.